

Kubernetes Selectors

What is a Selector in Kubernetes?

A Selector is a field used by a Service (and some other objects) to decide which Pods it should send traffic to.

Selector = a label-based filter to choose Pods.

Example:

Pods have labels like:

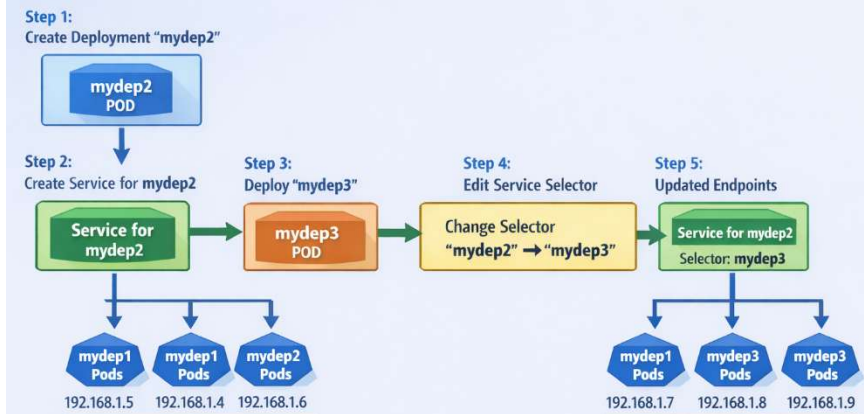
labels: app: mydep2

Service uses a selector like:

selector: app: mydep2

This means: “Attach this Service only to Pods where *app=mydep2*.”

Understanding Kubernetes Selectors



Create Deployment mydep2 (3 replicas)

Create a Deployment named mydep2 with 3 Pods.

Each Pod gets a label like:

labels: app: mydep2

```
controlplane:~$ kubectl create deployment mydep2 --image=docker.io/httpd --replicas=3
deployment.apps/mydep2 created
controlplane:~$ kubectl get pod -o wide
```

NAME	READY	STATUS	RESTARTS	AGE	IP	NODE	NOMINATED NODE	READINESS GATES
mydep2-585fddb96-bpxzg	1/1	Running	0	20s	192.168.1.5	node01	<none>	<none>
mydep2-585fddb96-dbt dx	1/1	Running	0	20s	192.168.1.4	node01	<none>	<none>
mydep2-585fddb96-qw4c5	1/1	Running	0	20s	192.168.1.6	node01	<none>	<none>

```
controlplane:~$ kubectl get pods --show-labels
```

NAME	READY	STATUS	RESTARTS	AGE	LABELS
mydep2-585fddb96-bpxzg	1/1	Running	0	32s	app=mydep2,pod-template-hash=585fddb96
mydep2-585fddb96-dbt dx	1/1	Running	0	32s	app=mydep2,pod-template-hash=585fddb96
mydep2-585fddb96-qw4c5	1/1	Running	0	32s	app=mydep2,pod-template-hash=585fddb96

Result:

Kubernetes creates 3 Pods, for example:

- Pod 1 → IP: 192.168.1.5
- Pod 2 → IP: 192.168.1.4
- Pod 3 → IP: 192.168.1.6

At this moment: These Pods are independent — no Service is attached yet.

Create a Service for mydep2

Now you create a Service from mydep2.

Service YAML (conceptually):

selector: app: mydep2

```

controlplane:~$ kubectl expose deployment mydep2 --port=80
service/mydep2 exposed
controlplane:~$ kubectl get pods --show-labels
NAME                                READY   STATUS    RESTARTS   AGE   LABELS
mydep2-585fddb96-bpxzg             1/1     Running   0           101s   app=mydep2,pod-template-hash=585fddb96
mydep2-585fddb96-dbt dx            1/1     Running   0           101s   app=mydep2,pod-template-hash=585fddb96
mydep2-585fddb96-qw4c5             1/1     Running   0           101s   app=mydep2,pod-template-hash=585fddb96
controlplane:~$ kubectl describe svc mydep2
Name:                               mydep2
Namespace:                         default
Labels:                            app=mydep2
Annotations:                        <none>
Selector:                          app=mydep2
Type:                              ClusterIP
IP Family Policy:                  SingleStack
IP Families:                       IPv4
IP:                                10.103.215.147
IPs:                               10.103.215.147
Port:                              <unset> 80/TCP
TargetPort:                        80/TCP
Endpoints:                        192.168.1.5:80,192.168.1.4:80,192.168.1.6:80
Session Affinity:                  None
Internal Traffic Policy:           Cluster
Events:                            <none>

```

What happens now?

The Service looks for all Pods with label `app=mydep2` and attaches to them.

Endpoints attached:

- 192.168.1.5
- 192.168.1.4
- 192.168.1.6

Concept cleared: The Service is not attached to Deployment — it is attached to Pods via Labels using Selectors.

Create Second Deployment mydep3 (3 replicas)

Now create another Deployment named mydep3.

Its Pods have a different label:

labels: app: mydep3

```

controlplane:~$ kubectl create deployment mydep3 --image=docker.io/nginx --replicas=3
error: failed to create deployment: deployments.apps "mydep3" already exists
controlplane:~$ kubectl get pod -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP              NODE     NOMINATED NODE   READINESS GATES
mydep2-585fddb96-bpxzg             1/1     Running   0           8m57s   192.168.1.5     node01   <none>            <none>
mydep2-585fddb96-dbt dx            1/1     Running   0           8m57s   192.168.1.4     node01   <none>            <none>
mydep2-585fddb96-qw4c5             1/1     Running   0           8m57s   192.168.1.6     node01   <none>            <none>
mydep3-678f5c5997-7n2gv           1/1     Running   0           5m23s   192.168.1.9     node01   <none>            <none>
mydep3-678f5c5997-pjtss           1/1     Running   0           5m23s   192.168.1.8     node01   <none>            <none>
mydep3-678f5c5997-w7wk6           1/1     Running   0           5m24s   192.168.1.7     node01   <none>            <none>
controlplane:~$

```

New Pods are created:

- Pod 1 → 192.168.1.7
- Pod 2 → 192.168.1.8
- Pod 3 → 192.168.1.9

At this moment:

- Service is still pointing to **mydep2**
- mydep3 Pods are not connected to any Service yet

Edit Existing Service mydep2

Edit the existing Service and change the selector from:

selector: **app: mydep2** to: selector: **app: mydep3**

```
controlplane:~$ kubectl edit svc mydep2
service/mydep2 edited
controlplane:~$ cat /tmp/kubectl-edit-2018033649.yaml
# Please edit the object below. Lines beginning with a '#' will be ignored,
# and an empty file will abort the edit. If an error occurs while saving this file will be
# reopened with the relevant failures.
#
apiVersion: v1
kind: Service
metadata:
  creationTimestamp: "2026-02-05T06:43:03Z"
  labels:
    app: mydep2
  name: mydep2
  namespace: default
  resourceVersion: "3384"
  uid: 57efbd3a-82a7-4ab4-8eb0-84002a32193c
spec:
  clusterIP: 10.103.215.147
  clusterIPs:
  - 10.103.215.147
  internalTrafficPolicy: Cluster
  ipFamilies:
  - IPv4
  ipFamilyPolicy: SingleStack
  ports:
  - port: 80
    protocol: TCP
    targetPort: 80
  selector:
    app: mydep2
  sessionAffinity: None
  type: ClusterIP
status:
  loadBalancer: {}
controlplane:~$
```

What does this mean?

You are telling Kubernetes:

“Stop selecting **mydep2** Pods, now select **mydep3** Pods.”

Check if Selector Changed

Run: `kubectl describe service mydep2`

You should see: Selector: `app=mydep3`

This confirms that the Service is now pointing to `mydep3`, not `mydep2`.

```
controlplane:~$ kubectl describe svc mydep2
Name:                mydep2
Namespace:           default
Labels:              app=mydep2
Annotations:         <none>
Selector:            app=mydep3
Type:               ClusterIP
IP Family Policy:    SingleStack
IP Families:         IPv4
IP:                 10.103.215.147
IPs:                10.103.215.147
Port:               <unset> 80/TCP
TargetPort:         80/TCP
Endpoints:          192.168.1.7:80,192.168.1.8:80,192.168.1.9:80
Session Affinity:    None
Internal Traffic Policy: Cluster
Events:             <none>
```

Observe the Endpoints Change

Old endpoints (before change):

- 192.168.1.5
- 192.168.1.4
- 192.168.1.6

New endpoints (after selector change):

- 192.168.1.7
- 192.168.1.8
- 192.168.1.9

```
controlplane:~$ kubectl get pods --show-labels
NAME          READY   STATUS    RESTARTS   AGE   LABELS
mydep2-585fddb96-bpxzg  1/1     Running   0          22m   app=mydep2,pod-template-hash=585fddb96
mydep2-585fddb96-dbtcd  1/1     Running   0          22m   app=mydep2,pod-template-hash=585fddb96
mydep2-585fddb96-qw4c5  1/1     Running   0          22m   app=mydep2,pod-template-hash=585fddb96
mydep3-678fdc5997-7nzcgv 1/1     Running   0          18m   app=mydep3,pod-template-hash=678fdc5997
mydep3-678fdc5997-pjtss  1/1     Running   0          18m   app=mydep3,pod-template-hash=678fdc5997
mydep3-678fdc5997-w7wk6  1/1     Running   0          18m   app=mydep3,pod-template-hash=678fdc5997
controlplane:~$ kubectl get pod -o wide
NAME          READY   STATUS    RESTARTS   AGE   IP          NODE          NOMINATED NODE   READINESS GATES
mydep2-585fddb96-bpxzg  1/1     Running   0          22m   192.168.1.5   node01        <none>           <none>
mydep2-585fddb96-dbtcd  1/1     Running   0          22m   192.168.1.4   node01        <none>           <none>
mydep2-585fddb96-qw4c5  1/1     Running   0          22m   192.168.1.6   node01        <none>           <none>
mydep3-678fdc5997-7nzcgv 1/1     Running   0          19m   192.168.1.9   node01        <none>           <none>
mydep3-678fdc5997-pjtss  1/1     Running   0          19m   192.168.1.8   node01        <none>           <none>
mydep3-678fdc5997-w7wk6  1/1     Running   0          19m   192.168.1.7   node01        <none>           <none>
```

Service does NOT care about Deployment names. It only cares about:
Labels + Selectors

Because am changed the selector from `app=mydep2` to `app=mydep3`, the Service automatically attached to new Pods.

Selectors decide which Pods a Service connects to based on labels — not Deployment names.